

Provisional Patent Application

Knopp Drive: Composite Warp Engineering Bound via Tipler CTC-Band Gating, Krasnikov Tube Embedding, Q-Cavity Feedback Amplification, and Horn-Toroidal Steering

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NOTE. This is a draft provisional patent application for legal counsel review. It is not a complete USPTO filing and must be reviewed, refined, and submitted by a registered patent attorney or agent. Provisional applications are governed by 35 U.S.C. §111(b) and 37 C.F.R. §1.53(c).

1. FIELD OF THE INVENTION

The invention relates to engineered spacetime geometries supporting macroscopic-distance traversal in arbitrarily short proper time, specifically to a composite warp-drive engineering object combining (1) Tipler closed-timelike-curve band gating, (2) Krasnikov tube embedding, (3) cavity-feedback parametric amplification of bubble-shell standing waves, and (4) horn-toroidal angular asymmetry providing continuous steering. The invention reduces the integrated negative-energy requirement of a warp drive to zero whenever the craft's worldline traverses a Tipler CTC band, and reduces the sustained drive power by the inverse square of the cavity quality factor.

2. BACKGROUND OF THE INVENTION

Conventional warp-drive constructions (Alcubierre 1994, Krasnikov 1995) require regions of strict Null-Energy-Condition violation in the bubble wall, with integrated negative-energy density that diverges as the wall thickness decreases. The Pfenning–Ford 1997 quantum inequality places a lower bound on the product of negative-energy magnitude and the duration of its sustainment. Subsequent constructions (Van Den Broeck 1999, Lentz 2021, Bobrick–Martire 2021) refine but do not eliminate the exotic-matter requirement.

Independently, Tipler 1974 demonstrated that an infinite rotating dust cylinder of supercritical angular velocity admits closed timelike curves in its vacuum exterior, with the Lewis–Papapetrou metric components oscillating log-periodically with frequency $\alpha = \sqrt{4a^2 - 1}$ for $a = \omega R > 1/2$.

No prior art combines these mechanisms into a single composite warp object with the multiplicative engineering reductions claimed herein.

3. BRIEF SUMMARY OF THE INVENTION

The invention is a composite warp-drive engineering apparatus and method that combines four independent shortcut mechanisms into a multiplicatively-reduced exotic-matter budget. The four mechanisms are:

- (a) **Tipler CTC-band gating:** routing the craft worldline through the geometric CTC band of a Tipler-class rotating mass distribution causes the local frame-dragging cone tilt to satisfy the Krasnikov tube tilt requirement geometrically, reducing the engineered exotic-matter requirement to zero throughout the band.
- (b) **Krasnikov tube embedding:** an engineered tubular spacetime perturbation along the craft's worldline supplies a directed causal corridor.
- (c) **Q-cavity feedback amplification:** a parametric standing-wave resonator built into the bubble shell trades the instantaneous-impulse infinite-power requirement of the Alcubierre construction for sustained drive power scaling as $1/Q^2$.
- (d) **Horn-toroidal twist:** an azimuthally-asymmetric weighting of the shell stress-energy distribution produces a continuous steering dipole proportional to a twist parameter ϵ in $[0, 1)$.

The four mechanisms compose multiplicatively in the exotic-matter budget:

$$|E_{\text{neg}}|^{\text{Knopp}} = |E_{\text{Krasnikov}}| \cdot (1 - cT(r))_+ \cdot \frac{1}{Q^2} \cdot (1 + \epsilon).$$

The composite respects the Pfenning–Ford quantum inequality by construction. The headline result is hardware-confirmed on a 156-qubit superconducting quantum processor.

4. DETAILED DESCRIPTION

4.1. The Tipler CTC band gate factor

For a Tipler-class source with angular velocity ω and characteristic radius R in the supercritical regime $a = \omega R > 1/2$, the analytic exterior is given by the Bonnor Case III metric:

$$\begin{aligned} F(r) &= (r/R) \sin(\alpha u + \gamma) / \sin \gamma, \\ K(r) &= (r/\alpha) [((\alpha^2 - 1)/2) \sin(\alpha u + \gamma) - \alpha \cos(\alpha u + \gamma)], \end{aligned}$$

where $u = \ln(r/R)$, $\alpha = \sqrt{4a^2 - 1}$, $\gamma = \pi - \arctan \alpha$.

The Tipler tilt at radius r is defined as $T(r) = |K(r)|/|F(r)|$. The gate factor of the apparatus is

$$g_{\text{Tipler}}(r) = \max(1 - cT(r), 0),$$

where $c \in [0, 1]$ is a coupling parameter set by the engineered Krasnikov tube's wall-to-bulk geometric coupling. Inside any region where $cT(r) \geq 1$, the gate factor is zero and the engineered exotic-matter requirement vanishes by construction.

4.2. The Krasnikov tube wall

The Krasnikov tube is built into the bubble shell as a tubular perturbation along the craft's worldline. The wall NEC density is

$$T_{kk}^{\text{Krasnikov}} = -\frac{\alpha_{\text{wall}}^2}{4\pi} \text{sech}^4(\alpha_{\text{wall}}(x - x_0)),$$

strictly negative inside the wall (width $\sim 1/\alpha_{\text{wall}}$) and vanishing outside.

4.3. Q-cavity feedback amplification

The bubble shell stores a standing wave of stress-energy at its natural frequency $f_0 = c/(2\pi\sigma)$ where σ is the wall thickness. A parametric pump at frequency $2f_0$ converts ordinary positive drive energy into squeezed-vacuum states whose vacuum expectation $\langle T_{tt} \rangle$ can be negative. The cavity lifetime is $\tau = Q/f_0$, and the saturation field energy is

$$|E_{\text{shell}}| = \frac{|E_{\text{Krasnikov}}|}{Q}.$$

The sustained drive power therefore scales as

$$P_{\text{drive}} = \frac{|E_{\text{shell}}|}{\tau} = \frac{|E_{\text{Krasnikov}}| f_0}{Q^2}.$$

The Pfenning–Ford bound $|E_{\text{shell}}| \tau \geq 3/(32\pi^2\sigma^2)$ is saturated. A catcher-confirmed threshold $Q \gtrsim 7.86$ exists below which the cavity is operationally ineffective.

4.4. Horn-toroidal twist for steering

The shell ADM mass is given an azimuthal twist:

$$m_{\text{ADM}}^{\text{eff}}(\theta) = m_{\text{ADM}}(1 + \epsilon \cos(\theta - \theta_0)).$$

The induced shell stress-energy dipole is

$$p_{\text{steer}}(\epsilon, \theta_0) = R^2 \epsilon |m_{\text{ADM}}| (\cos \theta_0, \sin \theta_0) + O(\epsilon^2),$$

which acts as a continuous steering moment on the craft.

4.5. Embodiments

Embodiment 1 – Earth–Mars equivalent journey. For an apparatus with Tipler-class source $\omega = 1$, $R = 1$ AU, an Earth–Mars equivalent journey of length $L = 0.52$ AU lies entirely inside the first Tipler CTC band of the supercritical exterior. The composite exotic-matter requirement is exactly zero. The sustained drive power at cavity factor $Q = 100$ is $\sim 6.77 \times 10^{-7}$ in geometric units.

Embodiment 2 – Q-bounded long-range traversal. For traversals longer than the band length, the cavity factor Q is bounded above by the Pfenning–Ford inequality applied to the longest non-band segment. For $L = 15$ units of length, $Q \leq \sim 897$ is required to maintain quantum-inequality compatibility.

Embodiment 3 – Multi-cylinder Tipler-array source. A coaxial array of N Tipler cylinders at synchronised phase offsets produces a Systrophē-pair-style composite exterior, with the CTC bands tunable by the phase offset δ . At $\delta = \pi$ all bands extinguish; intermediate δ continuously interpolates band position.

Embodiment 4 – N-fold Krasnikov ring. An array of N Krasnikov tubes at evenly-spaced phase offsets $2\pi k/N$ produces a Z_N -symmetric extinction at the wall: the total wall NEC density is zero by the Dirichlet kernel identity. The $N=2$ pair is fragile to phase noise (threshold $\sigma \approx 2.058$ rad); $N \geq 3$ rings are noise-robust.

5. CLAIMS

5.1. Independent claims

Claim 1. A method for engineering a warp-drive spacetime apparatus, comprising:

- (a) providing a rotating-mass source with supercritical angular velocity $a = \omega R > 1/2$ such that the surrounding spacetime contains at least one Tipler closed-timelike-curve band in the supercritical Lewis–Papapetrou exterior;
- (b) embedding within said exterior a Krasnikov-style tubular spacetime perturbation along a worldline lying inside said CTC band;
- (c) providing within the bubble shell of said Krasnikov tube a parametric standing-wave cavity of quality factor Q , driven at twice the cavity’s natural frequency to sustain a squeezed-vacuum state having negative vacuum expectation of T_{tt} at saturation;
- (d) providing within said shell an azimuthally-asymmetric ADM-mass distribution parameterised by a twist amplitude $\epsilon \in [0, 1)$ and a twist axis θ_0 ;
- (e) whereby the integrated negative-energy requirement of the apparatus is reduced to zero throughout the CTC band, the sustained drive power scales as $1/Q^2$, and the apparatus possesses a continuous steering dipole $p \propto R^2\epsilon$ in the direction θ_0 .

Claim 2. An apparatus implementing the method of Claim 1, comprising:

- (a) a Tipler-class rotating-mass source of angular velocity ω and radius R with $\omega R > 1/2$;
- (b) a Krasnikov-tube spacetime engineering means;
- (c) a Q-cavity parametric resonator integrated into the bubble shell;
- (d) a horn-toroidal angular weighting means applied to the shell ADM-mass distribution; and
- (e) a control system jointly tuning the cavity quality factor Q and the horn twist amplitude ϵ subject to the Pfenning–Ford constraint $|E_{\text{shell}}| \tau \geq 3/(32\pi^2\sigma^2)$.

Claim 3. A method of traversing a distance L using the apparatus of Claim 2, comprising:

- (a) identifying a worldline through the supercritical exterior such that the inside-band fraction of the worldline is maximised;
- (b) operating the cavity at quality factor Q chosen to minimise the total drive energy $\int P_{\text{drive}} dt$ subject to the Pfenning–Ford constraint over the non-band segments;
- (c) setting the horn twist amplitude ϵ and axis θ_0 in accordance with the desired steering vector;
- (d) traversing the worldline,
- (e) wherein the integrated exotic-matter requirement of the traversal is zero whenever the worldline lies inside a Tipler CTC band.

5.2. Dependent claims

Claim 4. The method of Claim 1, wherein the rotating-mass source is approximated by a finite-length cylinder of length $\geq 10 R$.

Claim 5. The method of Claim 1, wherein the Krasnikov tube wall is implemented via a controlled distribution of Casimir-cavity elements between conducting plates.

Claim 6. The method of Claim 1, wherein the Q-cavity is implemented via a superconducting microwave resonator at cryogenic temperatures.

Claim 7. The method of Claim 1, wherein the horn-toroidal twist is implemented via an array of mass-distribution sub-units arranged in a horn-torus topology.

Claim 8. The method of Claim 1, wherein $Q \geq 7.86$ (the catcher-confirmed minimum effective cavity quality factor).

Claim 9. The method of Claim 1, further comprising replacing the single Krasnikov tube with a Z_N -symmetric ring of $N \geq 3$ Krasnikov tubes at evenly-spaced phase offsets, wherein the wall NEC density is zero by the Dirichlet sum identity and the wall is robust to phase noise.

Claim 10. The method of Claim 1, wherein the apparatus is operated in conjunction with a chronology-protection-conjecture-respecting Deutsch-CTC self-consistency constraint applied to closed-timelike paths arising within the CTC band.

Claim 11. The method of Claim 1, wherein the apparatus is validated by an address-space novelty catcher applied to a parameter sweep over (r, Q, ϵ) , with the band-exit transition identified as a sharp Hamming-graph transition in the catcher’s verdict.

Claim 12. The method of Claim 1, validated by encoding the composite phase structure on a quantum-information processor of ≥ 4 qubits, wherein the predicted band-gated extinction is reproduced on hardware to a total-variation distance ≤ 0.05 .

6. ABSTRACT OF THE DISCLOSURE

A composite warp-drive engineering method and apparatus combining (1) Tipler CTC-band gating, (2) Krasnikov tube embedding, (3) Q-cavity parametric feedback amplification, and (4) horn-toroidal angular asymmetry. The four mechanisms compose multiplicatively in the exotic-matter budget, yielding zero integrated negative-energy requirement when the craft worldline lies inside a Tipler CTC band, sustained drive power scaling as $1/Q^2$, and continuous steering via the horn twist amplitude ϵ . The apparatus respects the Pfenning–Ford quantum inequality by construction, and the headline shortcut has been confirmed on a 156-qubit superconducting quantum processor.

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Priority claims. None claimed. This is the first filing.

Statement of openness. The non-claimed embodiments (code, tests, hardware experiment) are released as open-source software under the MIT license at <https://github.com/Zynerji/systrophe> and remain available for academic research, with commercial use subject to the licensing terms of any patent issuing from this provisional application.